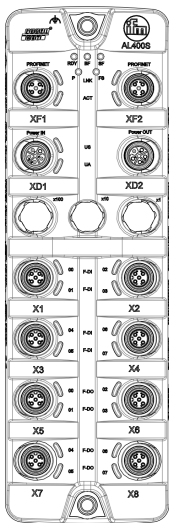


100_SRIO_Product-Requirements



Product Requirement Specifications



Project: 153829 Remote_IO_Safe_DIDO_PROFIsafe

Affected Products: eco310 / **AL400S** / PROFIsafe 6x2 F-DI 2x2 F-DO IP67
eco310 / **AL401S** / PROFIsafe 6x2 F-DI 2x2 F-DO IP69K

Status:	✔ Approved	Confidentiality:	🔒 Confidential
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Author:	Ruess, Bastian	Created:	2022-04-19
Review	✔ Budweiser, Markus (Quality) Signed 2026-06-18 07:54:53 Process-compliant review confirmed. The document is ready for release. (Reviewers are listed in the table 'Reviewers for current version.')		
Approval	✔ Engel, Sebastian (Technical Manager) Signed 2026-06-19 08:18:12		
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Approved Versions

Current Document version **V1.20** is **approved**.

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- **V1.16** (2023-12-04 13:33)
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Reviewers for current version

Info: This document is reviewed by using tool-based review process. Every workitem is reviewed in the document context by the R&D experts independently. Therefore the workitems are checked for correctness, completeness and consistence to other work-items and to the document context. The minimal number of independent reviewers are defined by the inno-process.

- [Mollnau, Stephan](#)
- [Riehm, Thomas](#)
- [Kaiser, Klaus](#)
- [Träger, Heiko](#)
- [Topalovic, Dragana](#)

✓ All workItems related with current version are approved

Change Log

(Typ: c = created, m = modified, + = added, - = deleted)

V1.00

Type	Chapter	Author	Description
c		S. Hummel	New document

V1.10

Type	Chapter	Author	Description
m	all	See Description	for document approval documentation see work item attachments

V1.11

Type	Chapter	Author	Description
m	all	A. Wiedekind-Klein (embeX)	solving of comments, update of item attribute

V1.12

Type	Chapter	Author	Description
m	5.6.1	D. Kunzelmann (embeX)	Omitted 2x 1 channel option for safe output in parametrization, now only 1x 2 channel or 1x 1 channel is possible
+	-	B.Ruess	Added table for review of document version regarding TÜV LOP reworks.
			for document approval documentation see work item attachments

V1.13

Type	Chapter	Author	Description
m	document head	Till Monzel	Changed document head according to process standard see: Rules for Safe_RIO team
m	SRIO-1206	Robert Vogt	description: added restart behaviour
			for document approval documentation see work item attachments

V1.14

Type	Chapter	Author	Description
m	document head	Till Monzel	Changed document head according to process standard see: Rules for Safe_RIO team
m	all	B.Ruess	Adjusted to new modified features.

V1.15

Type	Chapter	Author	Description
m	1.6 3 5.2 5.6	B.Ruess	Splitted norm reqs. to multiple single reqs. DI/DO port switch, LED "P" name change, adjusted Parameter according to new values.

V1.16

Type	Chapter	Author	Description
m	8,12	B.Ruess	Infomation workitem stated support of external tool which was skipped for this project. -> adjusted to current featureset. Adjusted SRIO-6579 and SRIO-6581.

V1.17

Type	Chapter	Author	Description
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m	all	B.Ruess	 SRIO-9309 - to be closed: Change Port Status LED to yellow  SRIO-9322 - Rework of ProduktRequirements based on TÜV Feedback  SRIO-9295 - Temperatur driven change of Portkonfiguration Those CRs were collected and are causing the changes below. The workitem are linked to the respective CR.
m	First page	B.Ruess	Changed Productname in HeadView to fit new naming.
-	previously 3 now 7	B.Ruess	Deleted remark/link to flowchart in SRIO-2013 since obsolete.
-	previously 2.2	B.Ruess	Moved Block diagramm in chapter 2.2 to  201_SRIO_System_Spec_Architecture
+	2.1	B.Ruess	Added Module Variants and picture of device
m	previously 3 now: 3,4,5,7,9	B.Ruess	Relocated requirements from chapter General product requirements in chapter HMI, Rotary switch, and IoT. Moved General product requirements behind Inputs&Outputs.
m	8.2	B.Ruess	Changed SRIO-9344 since the portnumbering and channels did not fit anymore.
-	all	B.Ruess	Deleted Requirements/Information Items reserved for DI-Only variant, since it won't be planned in future anymore.
+	6, 9	P.Beugel	Added Requirments regarding SELV and PELV in chapter General product requirements and Power Supply.  SRIO-9323 - SELV/PELV addition to ProductRequirements
m	6, 9	B.Ruess	Decision SELV is to be excluded all eco310 products. Change reverted (PELV is kept).  SRIO-11507 - The intended Use is PELV...(SELV is not required)
+	5	B.Ruess	Moved Requirements for IoT-Core from  130_Communication to this document.
m	all	B.Ruess	corrected the SIL Flag setting and Product names together with FSE/both FSM (A.Bellatreche, S.Mollnau)
m	-	B.Ruess	changed related requirements:  SRIO-15210 - tobeclosed - Standarts for SRIO  SRIO-14705 - SRIO-853 - DO Safe State  SRIO-14889 - Change on Requirement for Power Limitation  SRIO-14976 - Rewording Requirements based on Review 23.06.

V1.18

Type	Chapter	Author	Description
m	8.3	B.Ruess	SRIO-2097 moved from document since rejected.
m	16	B.Ruess	 SRIO-15742 - Rewording of EMC Requirement added for LED "[no supervision needed, impression of tester is sufficient]" for criterion A,B
m	1.6 + 6.1	B.Ruess	Updated Version  SRIO-17851 - New version 13849-1:2023

V1.19

Type	Chapter	Author	Description
m	all	B.Ruess	removed rejected work items from this document
m	1.6/1.7	B.Ruess	 SRIO-18309 - DNV excluded for current project -> rejected respective requirements.
m	1.6/15.1/15.2	B.Ruess	Updated Version:  SRIO-19181 - Update of the standard reference to IEC 61131-2:2017; Rejected  SRIO-7302 - Fire spread in case of a short circuit within the Device  SRIO-7303 - Clearance & creepage distance since duplicate of  SRIO-1781 ( SRIO-20689)
m	8.6.2	B.Ruess	Updated  SRIO-1157 - Parameters Safe Input based on  SRIO-19971 - iParameter FilterTime adjustment
a	16	B.Ruess	Created  SRIO-20173 &  SRIO-20174 based on  SRIO-20188 - Added requirements regarding EMC restrictions to cable connections
m	8.4	B.Ruess	 SRIO-724 - Diagnosis of Switching Components I adapted (splitted to two req.)

m	16.1	B.Ruess	EMC criterion(s) adapted to comply with company standards
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V1.20

Type	Chapter	Author	Description
m	8.4	B.Ruess	adapted Parameter/load definitions of DO config accordingly to  SRIO-22084 - Support of different Testpulse Duration and restructured chapter 8.4 as well as created  SRIO-22092 .

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1 Scope

1.1 Purpose of this document

This specification describes the product requirements for the "Safe remote I/O" project. The specification contains all functional and non-functional properties of the product. The requirements are taken from the document [IR] or derived, further detailed and completed.

Due to the project structure, the complete specification of the functional and non-functional properties of individual subcomponents is outsourced to separate component specifications.

The requirements specified in this document are the basis for the customer acceptance to be carried out at the end of the development. The specification must therefore be approved by the customer. Changes or extensions of the requirements after release require mutual agreement and will be recorded in a new revision of this document.

Safety Category: QM/SIL

For the product level the safety category is meant as signal that this workitem is important.

The correct usage in terms of functional safety will start from system level.

1.2 Document structure

The Product Requirement Specification (PRS) is split into several sub documents to keep the necessary overhead low, if changes are necessary only the appropriate sub-document need to be revised.

The documents are hosted within the "Polarion" software (ifm-environment) for revision management.

1.3 Referenced documents

Referenced documents (input documents) are documents that contain requirements or data that are reused, clarified or supplemented in this document. Changes in these documents can therefore have an influence on this document. Therefore, the specification of the date and/or version is mandatory for higher-level documents.

Table 1 Referenced Documents

Ref	Title	Baseline	Date
[IR]	Input Feature Specification  153829_SafeRemoteIO	V1.0	???
[ZVEI]	Position Paper CB24I Classification of Binary 24V Interfaces Functional Safety aspects covered by dynamic testing Positions_Papier_CB24I_Edition_2_0_1_Klassifizierung-binaerer-24-V-Schnittstellen.pdf (zvei.org)	2021.01	---

1.4 Naming Conventions

A number of abbreviations are used in these documents. The following list presents the most important abbreviations and their meaning.

[SRIO] – Safe Remote I/O- Module

[DO] – digital output

[DI] – digital input

[FSOE] - Fail Safe over EtherCAT

[PP] – positive power switching (Output "PP-switching")

[PM] – power ground (masse) switching (Output "PM-switching")

[TIA] - Profisafe-Master programming software (Siemens tool)

[JSON] - JavaScript Object Notation (moneoconfigure safe app)

[DIL] – Dual in line
[IPC] - Inter Processor Communication

1.5 Additional and Related Documents

Additional documents are documents that are part of this product requirements specification (PRS) but are outsourced to stand-alone documents for better clarity. Since the information workitem is referenced in the other documents, the PRS is mentioned for clarity.

Document	Version	Short Description
 100_SRIO_Product-Requirements	 SRIO-22085 - V1.20	Head Document for Product Requirements
 130_Communication	 SRIO-22160 - V1.18	Details requirements regarding fieldbus and IoT communication
 140_FW-Update	 SRIO-17016 - V1.16	Details requirements regarding the firmware update
 180_Housing	 SRIO-20618 - V1.16	Details requirements regarding the housing
 190_QualityRequirements	 SRIO-20828 - V1.11	Details requirements from service and quality departments

1.6 Norms and standards

SRIO-6517 - Norms and Standards with defined Version

In addition to the standards determined in this document, the defined standards (and specified versions) by FSM-Plan shall be applied.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-1777 - IEC 61508: SIL 3

The safe Remote IO has to comply to SIL3 according to IEC 61508:2010

Note: The achievable category is depending on user-configuration

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-1778 - EN 13849-1: Cat 4 / PL e

The safe Remote IO has to comply with category 4 PL e according to EN ISO 13849-1:2023.

Note: The achievable category and performance level is depending on user-configuration.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-6533 - Installation Description

The reached safety relevant values of EN ISO 13849 as well as IEC 61508 shall be made available to the customer, so that the customer is able to use the device in its own safety application.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-1776 - EC machinery directive 2006/42/EG

The safe Remote IO device counts to safety relevant elements in respect to annex IV of the EC machinery directive, and has to comply the Requirements of this Standard. The compliance of the EC machinery directive is assumed if the harmonized Standard EN ISO 13849-1:2023 is applied.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

Moved  [SRIO-1779 - The safe Remote IO shall comply with the EN 61326-3-1 to 16 - EMC Requirements](#)

Moved  [SRIO-1780 - The safe Remote IO-hardware shall comply with 61131-2 to 15.2 - Mechanical environmental](#)

Moved  [SRIO-7306 - The safe Remote IO-hardware shall comply with EN 61131-2 regarding the EMC resistance to 16 - EMC Requirements](#)

SRIO-7307 - The safe Remote IO-hardware shall comply with 61131-2 regarding the power supply

The safe Remote IO shall comply with the IEC 61131-2:2017 chapter 6.2.1.x .

Remark: this includes the following chapter:

6.2.1.1 Rated values and operating ranges

6.2.1.2 Improper equipment power supply connection

6.2.1.3 Voltage interruptions power ports

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

Moved  [SRIO-2086 - The safe Remote IO-hardware shall comply with EN 61131-6 regarding EMC to 16 - EMC Requirements](#)

Moved  **SRIO-7305** - The safe Remote IO-hardware shall comply with EN 61131-6 regarding mechanical test to 15.2 - Mechanical environmental

Moved  **SRIO-1781** - The safe Remote IO shall comply with the EN 61010-1 in respect to the following points to 15.2 - Mechanical environmental

Moved  **SRIO-7304** - Resistance of the identification plate to 15.2 - Mechanical environmental

Moved  **SRIO-7303** - Clearance & creepage distance to 15.2 - Mechanical environmental

Moved  **SRIO-7302** - Fire spread in case of a short circuit within the Device to 15.2 - Mechanical environmental

SRIO-1782 - Spec UL61010-2-201 shall be fulfilled

Spec **UL61010-2-201:2018** shall be fulfilled according electrical safety as well as applicable environmental and testing conditions.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-1783 - Spec UL 1310 shall be fulfilled

Spec **UL 1310:2019** shall be fulfilled

Class II shall be provided for connected devices.(The digital inputs and digital outputs act as a power source for connected devices)

 Must Have,  Approved,  AL400S,  AL401S,  QM

1.7 Required Approvals

SRIO-1787 - PROFINET Certification

A PROFINET Device certification shall be done. The PROFINET device conformance test shall be configured as followed:

1. Conformance Class: C
2. Net load Class: III

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-1870 - PROFIsafe Certification

A PROFIsafe Device (F-Slave) certification shall be achieved.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

2 Introduction and Overview

The overarching goal is the development of a product family of safe Remote- IO Devices [SRIO].
The product family might have DI-only, DO-only or mixed DI/DO Variants with different safety-protocols to either:

- transfer the input signal using the safety-protocol to,
- and/or switch the digital outputs based on the received information from the F-Host.

In general the device supports various Ethernet based fieldbus protocols, the fieldbus interface also enables the customer to connect via an IoT interface to the device for firmware updates or gather a insight in device information or statuses (only none safety related functionalities).

2.1 Module Variants

SRIO-1769 - Modules with PROFIsafe interface

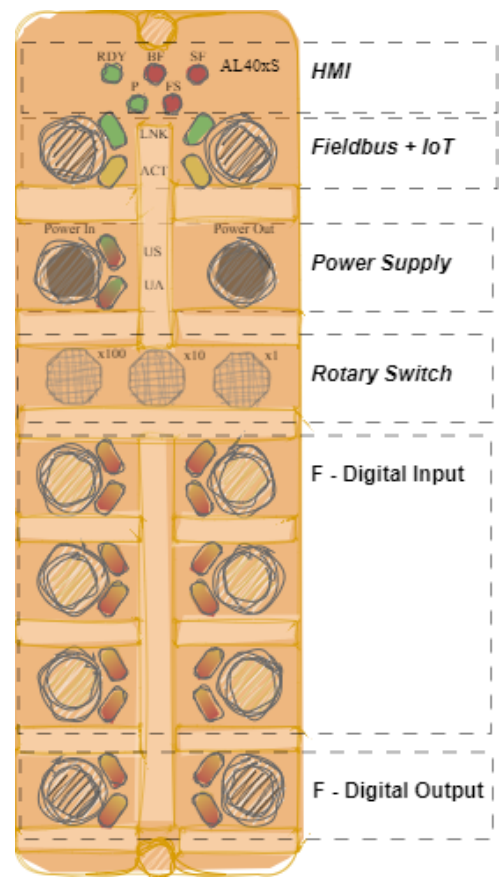
SRIO shall be developed as two variants as stated below.

Order ID Article description	IP	Fieldbus / Safety-Protocol	EA-System
AL400S ETH Module PROFIsafe 6DI 2DO IP67	IP67	PROFINET / PROFIsafe	6x2 F-DI and 2x2 F-DO
AL401S ETH Module PROFIsafe 6DI 2DO IP69K	IP69K	PROFINET / PROFIsafe	6x2 F-DI and 2x2 F-DO

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-9324 - ETH Module 6x2 F-DI and 2x2 F-DO

Visualization of the final product:



 Must Have,  Approved,  AL400S,  AL401S,  QM

3 HMI

SRIO-6531 - RDY LED

The device shall have an LED (green/red) signaling the functionality of the Host CPU.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6530 - BF LED

The device shall have an LED (green/red) signaling bus failure.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6529 - SF LED

The device shall have an LED (green/red) signaling system failure.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6528 - P LED

The device shall have an LED (green/red) signaling the state of the safe communication.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6527 - FS LED

The device shall have an LED (red) signaling the safe state of the device.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6526 - LNK LEDs

The device shall have LEDs (green) signaling the LINK for the fieldbus ports.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6525 - ACT LEDs

The device shall have LEDs (yellow) signaling the communication speed (10/100 MBit/s) of the fieldbus ports.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6524 - US LED

The device shall have an LED (green/red) signaling the state of the sensor supply US.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6523 - UA LED

The device shall have an LED (green/red) signaling the state of the actuator supply UA.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6532 - Port Channel LEDs

For each port channel, the device shall have an LED (yellow/red).

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

4 Fieldbus

SRIO-1768 - NetX90 for fieldbus communication

The ethernet-interface integration shall be done on the Basis of the NetX90 for the fieldbus-communication

 Must Have,  Approved,  AL400S,  AL401S,  QM

Fieldbus related definitions are located in  [130_Communication](#)

5 IoT / Webserver (ifm IoT-Core)

SRIO-2100 - Webserver ifm IoT Core

The device shall have the ifm IoT Core integrated which is used as web interface.

Please refer to Webserver Features for a specification of the features that the device shall provide.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

IoTCore related definitions are located in  [130_Communication](#)

6 Power

6.1 Power Supply

Power Supply is 24V and split into two power domains Sensor supply (US) and Actuator supply (UA), whereby US is for input signals and internal circuits and UA is reserved for outputs only.

SRIO-683 - Supply Range

The product must be operable with a nominal supply voltage of 24VDC, with a range of +25% -20% based on the nominal value [19,2 V - 30 V].

⚠️ Should Have, ✅ Approved, ➡️ SRIO platform, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-699 - Power Domains

The device shall have two power supplies US and UA.

⚠️ Must Have, ✅ Approved, ➡️ SRIO platform, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-1203 - Power Domains - Galvanic Separation

The two power domains of the device (US and UA) must be galvanic separated.

The isolation must comply with reinforced or double insulation up to 60V.

The isolation must have the fault exclusion acc. to EN ISO 13849-1:2023 and EN 61800-5-2:2017.

⚠️ Must Have, ✅ Approved, ➡️ SRIO platform, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-796 - Power Based Limitation

An power based current-limitation must be implemented for both supply voltages.

The power consumption must not exceed 100W (UL1310-Class II) in a single fault condition.

⚠️ Must Have, ✅ Approved, ➡️ SRIO platform, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-1775 - UA Current Limitation

It shall be possible to draw 2A for each DO-Port (splitted via both Channel).

⚠️ Should Have, ✅ Approved, ➡️ SRIO platform, 📋 AL401S, 📋 AL400S, 📋 QM

SRIO-8624 - External Power Supply of SRIO

SRIO shall be able to be powered external by PELV.

⚠️ Must Have, ✅ Approved, ➡️ SRIO platform, 📋 AL400S, 📋 AL401S, 📋 QM

6.2 Power Connectors

The connection of the power domains will be done by M12 connectors, the characteristics of the connectors are specified in this chapter. The connectors are called XD1 and XD2, whereby XD1 is the input, XD2 is the output.

SRIO-698 - Power Connectors

The product must have two power connectors, one input and one output (daisy chain)

⚠️ Must Have, ✅ Approved, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-702 - Power Input Connector

The power input connector must be of type M12, L-coded and male.

⚠️ Must Have, ✅ Approved, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-701 - Power Output Connector

The power output connector must be of type M12, L-coded and female.

⚠️ Must Have, ✅ Approved, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-1772 - The L-Coded Connectors shall be rated for use up to 105°C

The L-Coded Connectors shall be rated for use up to 105°C

⚠️ Must Have, ✅ Approved, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-1773 - The L-Coded Connectors shall carry a current of 9 A per Pin at 60°C, 16 A at 45°C

The L-Coded Connectors shall carry a current of 9 A per Pin at 60°C ambient, 16 A at 45°C ambient (between 45° and 60° a derating of the current is acceptable)

⚠️ Must Have, ✅ Approved, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-695 - Functional Earth

The device must have two mounting lugs. FE-Pin of the power connectors shall be connected to the top mounting lug.

⚠️ Must Have, ✅ Approved, 📋 AL400S, 📋 AL401S, 📋 QM

SRIO-11020 - Connector Pinout

The power connectors must have the following pinout:

Electrical connection					
Power Supply Port					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	L+ (US)	L+	Supply	US	24 V DC (US)
2	L- (UA)	L-	GND	UA	0 V DC (UA)
3	L- (US)	L-	GND	US	0 V DC (US)
4	L+ (UA)	L+	Supply	UA	24 V DC (UA)
5	FE	FE	FE	FE	Connection to Functional Earth

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

7 Rotary switch

SRIO-1347 - Rotary Switch Definition

As seen in the figure below, the device shall have 3 rotary switches, each being adjustable between 0 and 9.

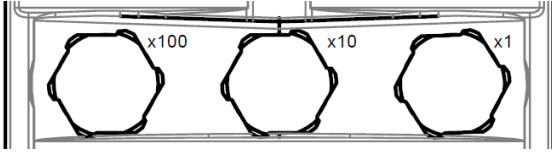


Figure 1 Definition of the rotary switches (covered)

The resulting number shall be the sum of the rotary switches 0 to 2 multiplied by their respective multiplier.

❗ Must Have, ✅ Approved, 📄 AL400S, 📄 AL401S, 📄 QM

SRIO-1344 - Delivery State of Rotary Switch

The delivery state of the rotary switch shall be 0.

❗ Must Have, ✅ Approved, 📄 AL400S, 📄 AL401S, 📄 QM

SRIO-2013 - Rotary Switch Functions

The rotary switches value shall be interpreted as follows:

- 0: delivery state of the rotary switches
- 1...899: own safe address
- 900...998: reserved for future functionalities
- 999: Firmware Update. The product shall boot in the update mode and shall accept the firmware update request via IoTCore.

❗ Must Have, ✅ Approved, ➡ SRIO platform, 📄 AL400S, 📄 AL401S, 📄 QM

8 Input & Output

8.1 Numbering of Ports and DIs/DOs

For ports and DIs/DOs the following numbering is applied:

1. Port Input and Output connections of the SRIO are called ports.

The SRIO has 8 of these ports, each connected to an M12-connector.

A port can be an input or an output port, depending on the SRIO-version. The name of one port is: Port n (n: X1..X8).

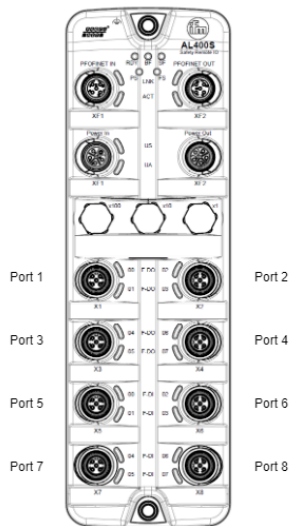


Figure 2 DI/DO Port Number

2. Channels (port-related numbering).

Each port has two channels:

- Port-channel 1 and Port-channel 2
- Globally they will be counted continuously as long as the port continues the same functionality:
e.g. Port 1 and Port 2 are Digital Inputs, therefore Port 1 has Channel 1 and 2 and Port 2 has Channel 3 and 4
- The mapping of port-channel 1 and port-channel 2 to the respective connector pin is shown in ☐ ~~SRIO-719~~ - Missing cross-reference and in ☐ ~~SRIO-717~~ - Missing cross-reference.

8.2 DI/DO Number Mapping

A mapping is needed between a given DI/DO number and which connector at the hardware level is denoted by this DI/DO number. In this chapter the DI/DO number mapping is described.

SRIO-2060 - DI/DO Numbering, DI/DO Device

For a DI/DO device the DI/DO numbers shall denote the port and port-channels as follows:

DI/DO number	Port	Port Pin	Function
DI 1	1	4	Channel 0
DI 2	1	2	Channel 1
DI 3	2	4	Channel 2
DI 4	2	2	Channel 3
DI 5	3	4	Channel 4
DI 6	3	2	Channel 5
DI 7	4	4	Channel 6

DI 8	4	2	Channel 7
DI 9	5	4	Channel 8
DI 10	5	2	Channel 9
DI 11	6	4	Channel 10
DI 12	6	2	Channel 11
DO 1	7	4	Channel 12
DO 2	7	2 (P/PP) or 3 (PM)	Channel 13
DO 3	8	4	Channel 14
DO 4	8	2 (P/PP) or 3 (PM)	Channel 15

Figure 3 : DI/DO numbers for 6x2 F-DI and 2x2 F-DO

Note: The Port Pinning is defined in  [SRIO-1990 - IO Connectors](#)

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

Note: The DI/DO number mapping cannot be changed via parameterization. The parameter "Safe Data Mapping" (see [13.2.3 - Parameter: "Safe Data Mapping"](#)) only determines how the data is represented in the process data, but it does not change the number mapping, i.e. which port|port-channel is denoted by a given DI/DO number.

SRIO-9842 - Cable Length for IO

The usage of the device is restricted to cable length for input and output ports to less or equal 30m.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-12749 - Cable Installation

The cable installation for SRIO for input and output ports shall be indoor installation only.

 Must Have,  Approved,  AL400S,  AL401S,  QM

8.3 F-Digital Inputs

Digital Inputs are meant to connect various kinds of sensors. In order to support failure detection, these inputs must provide special features like a switchable sensor supply.

The block circuit gives an overview about the structure of such an digital input.

Remark: This diagrams contain a first draft of the design. Changes may become necessary in the course of the development process.

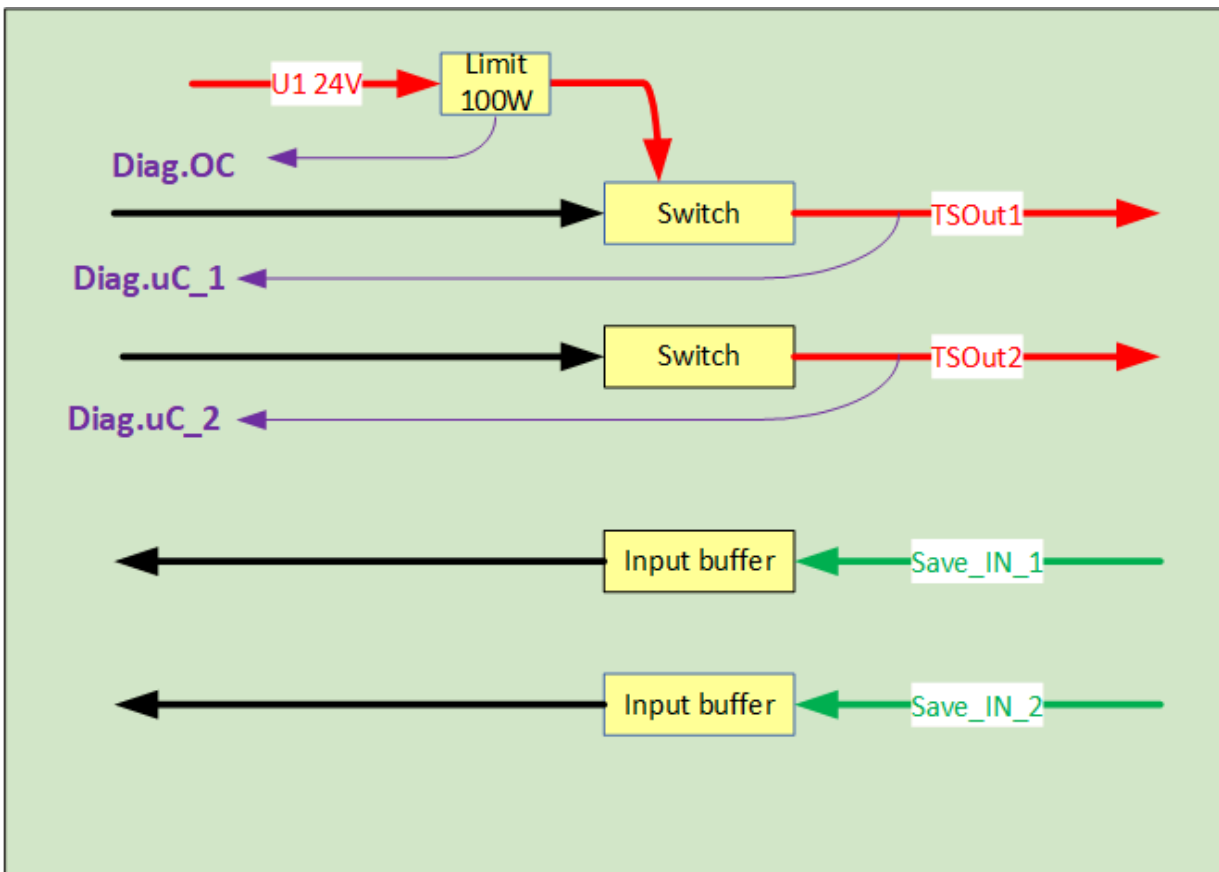


Figure 4 : Block diagram of digital input

SRIO-774 - Input Characteristic

The input characteristic of the safe Inputs shall follow the IEC 61131-2:2017 Type 3

⚠ Must Have, ✓ Approved, 📄 AL400S, 📄 AL401S, 📄 QM

SRIO-798 - Sensor Supply

Each digital input port must provide a sensor supply (based on power domain US)

⚠ Must Have, ✓ Approved, ➡ SRIO platform, 📄 AL400S, 📄 AL401S, 📄 QM

SRIO-799 - Switchable Sensorsupply

In case of Sensor Type C, the power of the sensor supply must be switchable.

⚠ Must Have, ✓ Approved, ➡ SRIO platform, 📄 AL400S, 📄 AL401S, 📄 SIL 3

SRIO-797 - Over-current Error

In case of an over-current at a port, the affected port shall be switched off and a diagnosis message shall be sent to the fieldbus system.

⚠ Must Have, ✓ Approved, ➡ SRIO platform, 📄 AL400S, 📄 AL401S, 📄 QM

SRIO-854 - DI Safe State

In the safe state the inputs of the device must report an inactive state.

⚠ Must Have, ✓ Approved, 📄 AL400S, 📄 AL401S, 📄 SIL 3

SRIO-1946 - Power-Consumption of the device shall be able to read out

The Power-Consumption of the device should be measured and data available for diagnostic purposes (Web-Server).

⚠ Should Have, ✓ Approved, 📄 AL400S, 📄 AL401S, 📄 QM

SRIO-2098 - Capacitive Load on Sensor-Supply

The supply of the inputs shall support capacitive load of: Cable length 30 m + 20 nF

⚠ Must Have, ✓ Approved, ➡ SRIO platform, 📄 AL400S, 📄 AL401S, 📄 QM

SRIO-6569 - Input Port Usage

The device shall support 1-channel and 2-channel configuration of the input ports.

⚠ Must Have, ✓ Approved, ➡ SRIO platform, 📄 AL400S, 📄 AL401S, 📄 SIL 3

SRIO-1963 - Achievable SIL/PL depending on Parametrization of SRIO and external Sensor

According to the input port usage, the following safety levels must be achieved:

Input Type [ZVEI]	Safety Performance	Timing
Type A, 2-channel usage	SIL3 / PLe cat. 4	see  SRIO-1967 - Timings - Input - Max Response Time SRIO
Type A, 1-channel usage	SIL2 / PLd cat. 2	see  SRIO-1967 - Timings - Input - Max Response Time SRIO
Type C (sensor with self-test feature), 2-channel equivalent/ antivalent usage	SIL3 / PLe cat. 4	see  SRIO-1967 - Timings - Input - Max Response Time SRIO
Type C, 1-channel usage	SIL2 / PLd cat. 2	see  SRIO-1967 - Timings - Input - Max Response Time SRIO
passive sensor (sensor without self-test feature) 2-channel equivalent/ antivalent usage with protected wiring	SIL3 / PLe cat. 4	see  SRIO-1967 - Timings - Input - Max Response Time SRIO
Passive Sensor (sensor without self-test feature), 2-channel equivalent/ antivalent usage	SIL2 / PLd cat. 3	see  SRIO-1967 - Timings - Input - Max Response Time SRIO

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

8.4 F-Digital Outputs

Digital outputs are meant to connect various kinds of actuators. In order to support failure detection these outputs must provide special features like a short circuit protection and overload detection or fail-safe modes in case of an microprocessor issue. In order to work as safe outputs, a watchdog hardware control must be added.

The block circuit gives an overview about the structure of such an digital output:

Remark: This diagrams contain a first draft of the design. Changes may become necessary in the course of the development process.

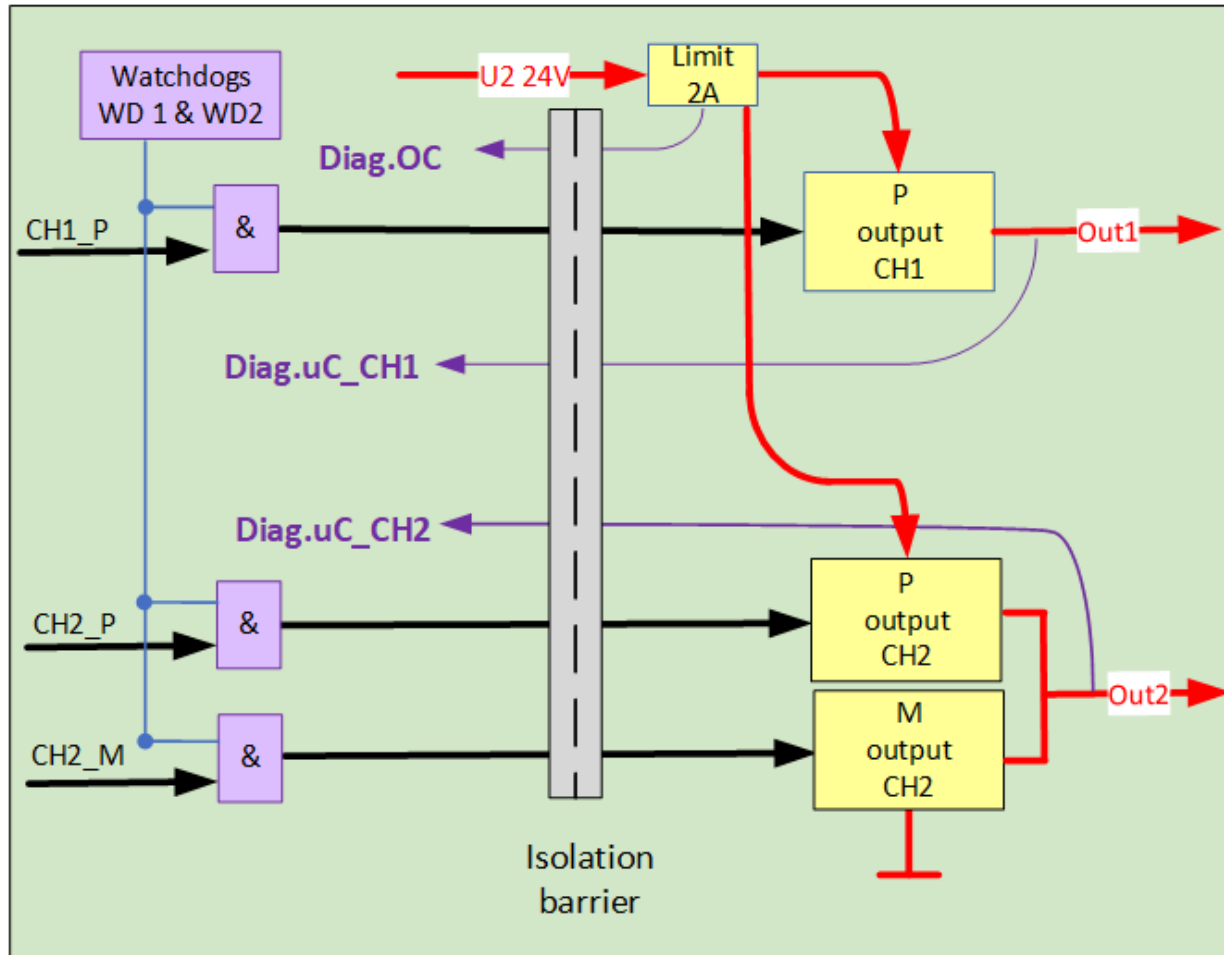


Figure 5 : Blockcircuit of digital output

8.4.1 Features

SRIO-7557 - Output Port Usage

The device shall support 1-channel and 2-channel configuration of the output ports.

⚠ Must Have, ✅ Approved, ➡ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

SRIO-7556 - Achievable SIL/PL depending on Parametrization of SRIO and external Actuator

According to the output port usage, the following safety levels must be achieved:

Output Type [ZVEI]	Safety Performance	Timing
Type C, PP switching, 1-channel usage	SIL2 / PL d / cat. 3	see 📄 SRIO-1965 - Timings - Output - Max Response Time SRIO
Type C, PP switching, 2-channel usage	SIL3 / PL e / cat. 4	see 📄 SRIO-1965 - Timings - Output - Max Response Time SRIO
Type D, PM switching, 2-channel usage	SIL 3 / PL e / cat. 4	see 📄 SRIO-1965 - Timings - Output - Max Response Time SRIO

⚠ Must Have, ✅ Approved, ➡ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

SRIO-851 - Switching Mode

For each output PP-Switching and PM-Switching must be possible.

❗ Must Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-852 - Switching Mode Selection

The switching mode PP or PM must be selectable by parameter.

Selection of the mode during parametrization only, not during operate.

❗ Must Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-1086 - Single Output failure

In case of an error at a digital output is detected, the affected digital output shall be switched off (Single output failure).

Error examples:

- over-current
- short circuit
- failure during output test

The state of each output can be checked by the Fieldbus-System.

❗ Must Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

SRIO-721 - Overload-Protection

Each digital output must have an overcurrent-protection. For the error reaction see 📄 [SRIO-1086 - Single Output failure](#).

❗ Must Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

SRIO-1206 - Safety Critical Failure

To reactivate a switched-off output (see 📄 [SRIO-1086 - Single Output failure](#)), it must first be acknowledged.

Note:

If all diagnosis are passed and an acknowledge was sent the Output will be able to be turned on.

The restart interlock is limited to the functionality described here. If a restart is safety-critical, it must also be handled or prevented at the system level (e.g. in the PLC) using appropriate measures.

❗ Must Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

8.4.2 Load Definitions

SRIO-730 - UA Current-Limitation per port

Each output port shall provide 2A@24V as current sum.

Note: According EN61131-2 the output must be capable to supply twice of the nominal current for 10ms.

⚠ Should Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-22114 - Switching Frequency DO

The outputs shall support a switching frequency of up to 10 Hz.

❗ Must Have, ✅ Approved, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-1092 - Inductive Loads on Output

The digital outputs shall support inductive loads up to 1 H.

❗ Must Have, ✅ Approved, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-857 - Inductive Load

It should be possible for each output port to drive inductive loads of category DC13 with 20W according to IEC 60947-5-1.

❗ Must Have, ✅ Approved, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-855 - No external fly back diode

An external fly back diode should not be required for DC-13 load (60947-5-1).

A suppressor diode with ± 36 V should be implemented.

Remark: The maximum inductive and capacitive load must be specified in the SRIO manual. In addition, the maximum switching frequency must be specified for the loads (depending on the load).

⚠ Should Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-856 - Connected Input Devices

All digital outputs should support input-devices with a Typ-2-characteristic according to IEC 61131-2.

⚠ Should Have, ✅ Approved, ⚠ AL400S, ⚠ AL401S, 📘 QM

SRIO-6515 - Discharge Shall Not Interrupt Safety Function

Discharging of external loads is not a safety relevant function. However, errors in the discharge circuit must not, disrupt the safety function.

❗ Must Have, ✅ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

8.4.3 Testpulse

SRIO-2087 - Switch-off capability test of the P and M branches

A switch-off capability test of the P and M branches of the safe outputs shall be integrated

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-22092 - Supported Testpulse (Switch-off Capability test)

SRIO shall support the following switch-off testpulse timings and respective loads:

Testpulse	Ohmic loads	Capacitive Load C_L	Inductive loads I_L
0.5 ms	2 A	1 μ F	0.5 H
1 ms	2 A	10 μ F	1 H
3 ms	2 A	10 μ F	1 H

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-724 - Diagnosis of Switching Components I

The system shall provide a parameter that enables OR disables the switch-off test pulses for each output port.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-21925 - Diagnosis of Switching Components I

When test pulses for digital outputs are disabled, the system shall require the user to ensure that each output is switched within a certain time limit [e.g. 8h]; this constraint shall be documented in the Safety Manual.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-1968 - Diagnosis of Switching Components II

The safety manual must describe, which measures must be taken when the switch-off test is disabled (see  SRIO-724 - Diagnosis of Switching Components I).

For example: The outputs must be tested at least every 8 hours by a switching.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

8.4.4 General

SRIO-708 - Power Supply for Digital Outputs

Each digital Output must be powered by UA power domain.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-1085 - Applicable Cable Length

The applicable cable length for the outputs must be ≤ 30 m

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-853 - DO Safe State

In the safe state, channels shall be set to Low.

Note:

In the case of "P" the inactive channel is always in safe state, therefore "P" and "PP" are the same:

"P" / "PP": both high side switches of a port are switched to open

"PM": both high side switches of a port are switched to open and also the low side switch is open.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

8.5 IO Connectors

The connection of the inputs and outputs will be done by M12 connectors, the characteristics of the connectors are specified in this chapter.

8.5.1 Input

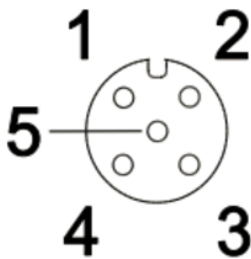
SRIO-718 - Digital Input Connector

The digital input connector must be of type M12, A-Coded, 5 pins and **female**.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-12541 - Input Connector Pinout

The digital input connector must have the following pinout:



Electrical connection					
Safety Input Port					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	L+ (US)/ TSOu1	L+	Supply F-DI1	US	24 V DC (US) Testsignal / Supply related to F-DI1
2	F-DI2	F-DI	F-DI2	US	Safety Digital Input 2
3	L- (US)	L-	GND	US	0 V DC (US)
4	F-DI1	F-DI	F-DI1	US	Safety Digital Input 1
5	L+ (US)/ TSOu2	L+	Supply F-DI2	US	24 V DC (US) Testsignal / Supply related to F-DI2

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

8.5.2 Output

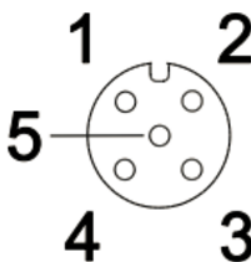
SRIO-716 - Digital Output Connector

The digital output connector must be of type M12, A-Coded, 5 pins and **female**.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-11008 - Digital Output Connector Pinout

The digital output connector shall have the following pinout:



Electrical connection					
Safety Output Port PP-Switching					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	n.c.	n.c.	-	-	-
2	F-DO2	F-DO	F-DO2	UA	Safety Digital Output 2
3	L- (UA)	L-	GND	UA	0 V DC (UA)
4	F-DO1	F-DO	F-DO1	UA	Safety Digital Output 1
5	FE	FE	FE	FE	Connection to Functional Earth

Electrical connection					
Safety Output Port PM-Switching					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	n.c.	n.c.	-	-	-
2	n.c.	n.c.	-	-	-
3	L- (UA)	L-	GND	UA	0 V DC (UA)
4	F-DO1	F-DO	F-DO1	UA	Safety Digital Output 1
5	FE	FE	FE	FE	Connection to Functional Earth

 Must Have,
  Approved,
  SRIO platform,
  AL400S,
  AL401S,
  QM

8.6 Binary Interfaces

8.6.1 Source (Outputs)

SRIO-1252 - Parameters Safe Output

Each output port shall have the following parameterization possibilities:

Table 2 Parameters Safe Output

Parameter-Name	Description	Parameter-Range (default is bold)
F-DO 1	Enables the first output of the port.	Enabled Disabled
F-DO 2	Enables the second output of the port.	Enabled Disabled
Testpulse	Defines if the testpulse shall be activated.	Enabled Disabled
Testpulse Duration	Defines the testpulse duration. (available for user information)	0,5 ms 1 ms 3 ms
Symmetry	Defines if the signals are processed individually (1oo1) or in combination to each other (1oo2).	1oo1 P 1oo2 PP 1oo2 PM
Switch-OFF Delay 1	Defines a switch-off Delay of the first output of the port.	Disabled, 250 ms, 500 ms, 1000 ms, 2000 ms, 4000 ms, 8000 ms
Switch-OFF Delay 2	Defines a switch-off Delay of the second output of the port.	Disabled, 250 ms, 500 ms, 1000 ms, 2000 ms, 4000 ms, 8000 ms

❗ Must Have, ✅ Approved, 📄 AL400S, 📄 AL401S, 💰 SIL 3

SRIO-1089 - Interface Types (Source)

Type C according to ZVEI; Positionspapier CB24I must be supported by each output port.

❗ Must Have, ✅ Approved, 📄 AL400S, 📄 AL401S, 💰 SIL 3

SRIO-1247 - Interface Types (Source)

Type D according to ZVEI; Positionspapier CB24I must be supported by each output port.

❗ Must Have, ✅ Approved, 📄 AL400S, 📄 AL401S, 💰 SIL 3

8.6.2 Sink (Inputs)

SRIO-1955 - Type A Input

Each input port shall be configurable as a Type A input according to the ZVEI-paper [ZVEI].

❗ Must Have, ✅ Approved, ➡ SRIO platform, 📄 AL400S, 📄 AL401S, 💰 SIL 3

SRIO-1956 - Type C Input

Each input port shall be configurable as a Type C input according to the ZVEI-paper [ZVEI].

❏ Must Have, ✔ Approved, ↔ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

SRIO-1157 - Parameters Safe Input

Each input port shall have the following parametrization possibilities, defaults are highlighted:

Table 3 Parameters Safe Input

Parameter-Name	Description	Parameter-Range (default is bold)
F-DI 1	Enables the first input of the respective port.	Enabled Disabled
F-DI 2	Enables the second input of the respective port.	Enabled Disabled
Testpulse Supply 1	Enables the supply belonging to the first input of the respective port.	Enabled Disabled
Testpulse Supply 2	Enables the supply belonging to the second input of the respective port.	Enabled Disabled
Symmetry	Defines if the signals are processed individually (1oo1) or in combination to each other (1oo2).	1oo1 1oo2 equivalent 1oo2 antivalent
Symmetry Discrepancy Time	In case of symmetry, the symmetry discrepancy time setting defines the timespan in which the symmetry definition shall be achieved.	Not analysed 10ms, 20ms, 50ms, 100ms, 500ms, 1000ms, 5000ms,
Resolve Symmetry Violation	Defines, in case of a symmetry violation, if a test-0-signal shall be necessary or if an acknowledgment is sufficient.	Acknowledgment sufficient Switch-on-inhibit
Filter F-DI 1	Enables the filter for first input.	5ms 10ms
Filter F-DI 2	Enables the filter for second input.	5ms 10ms

❏ Should Have, ✔ Approved, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

9 General product requirements

For product development, there are a number of general requirements that are not assigned to specific measurable technical characteristics. Other features are to be regarded as generally favorable and thus valid for many sub-points.

For the sake of clarity, these are summarized in the following chapter.

Some specific requirements that are assigned directly to functional units (e.g. housing, firmware, operating system) are described in separate documents. (see chapter [1.5 - Additional and Related Documents](#) of this document)

SRIO-8622 - Protection Class

SRIO shall cover the requirements according to protection class III, PELV, DIN EN 61140 VDE0140-1:2016-11.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-1793 - Processor STM32F777

The processor STM32F777 shall be used for the host and for both safety controllers.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-1771 - Ethernet Interface speed must support 100Mbps

The Ethernet interface must support: 100BASE-TX according to IEEE-802.3u

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-1083 - Voltage and Current Accuracy

The accuracy of voltage- and current-measurements should be $\leq 10\%$ based on absolute value.

Remark: Usage only for Webserver (maintenance purpose); Measurements in the context of this requirement are not safety related

 Should Have,  Approved,  AL400S,  AL401S,  QM

SRIO-2580 - Debug Interface

The device shall have an interface for debugging and firmware flashing for each controller.

This interface shall only be useable during development phase.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-1526 - Test Interface

The device shall have a test interface, enabling to trigger SW-FITs or receiving error messages.

This interface shall only be useable during development phase.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-1272 - Non-safe Code Partitions

On the safety controllers it shall be possible to add non-safe partitions of code and non-safe partitions of memory.

Note 1: The safe μ COS-II provides mechanisms to implement the freedom of feedback between safe and non-safe partitions.

A HowTo-description shall be done covering the following topics:

1. where in the design/code a non-safe task should be added
2. how non-safe data must be configured/declared
3. if necessary: requirements that the non-safe task must fulfill

This description shall be part of the Software Architecture Description ( [40_500_Software_Architecture](#)).

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-1268 - Safe Firmware Architecture Characteristics

The safe firmware architecture, design and code shall have the following characteristics:

1. modular design (= no monolithic construct)
2. layering design with the idea that higher layers use the lower layers for accomplishing their tasks
3. it shall be possible to add more inputs and more outputs without necessity to change the complete design

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-1262 moved to system level, 200_SRIO_System-Architecture chapter 7.2 Subsystem Safety

SRIO-1260 - Operating System (OS)

The operating system μ COS-II from Micrium shall be used. Safe uCOS-II shall be used on the Safe CPU's, Non-Safe VHIP3 uCOS-II on the Host-Controller.

Note: The safe μ COS-II is specified in  [150_μCOS-II_with_MPU](#)

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-1258 - EEPROM Layout

The EEPROM layout of the **safe** EEPROMs shall be according to the specification in *VHPP/BuildingBlocks/Device Information*. The parts of *VHPP/BuildingBlocks/Device Information* that are not needed / not applicable for the safe controllers may be implemented differently or do not need to be implemented. Note: The EEPROM of VHIP is covered by the VHIP specification and implementation.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-6572 - AIDA Position Paper

The device should apply the AIDA requirements for Safe Remote IOs.

 Should Have,  Approved,  AL400S,  AL401S,  QM

9.1 Declarations

SRIO-6574 - CE Declaration

The device shall meet all applicable safety, health, and environmental protection requirements which the European Union defined as necessary to apply the CE marking.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-6573 - UKCA

The device shall meet all applicable safety, health, and environmental protection requirements which the United Kingdom defined as necessary to apply the UKCA marking.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-6576 - Risk analysis

A risk analysis shall be performed.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

9.1.1 FCC

SRIO-11352 - FCC and ICES Conformity

FCC Conformity Class A shall be reached by SRIO.

 Must Have,  Approved,  AL400S,  AL401S,  QM

10 Operating Modes

Initially Operating Modes were planned to handle the systemstates and possible subsystem states.

As shown by [11 - Operating States](#) there are only states left which are also incorporated by the subsystems.

This chapter is therefore no longer valid and only kept for traceability (rejected requirements) purpose.

11 Operating States

From a whitebox point of view safe RIO is composed of two subsystems: subsystem communication part and subsystem safety.

For the Safe RemoteIO there are the following states which are also incorporated by the two subsystems which can have the same or less states:

1. *INIT*
2. *NO_PARAM*
3. *OPERATING*
4. *FW_UPDATE*
5. *FATAL_ERROR*

11.1.1 Init State

SRIO-1417 - Init State

There shall be an init state which shall do the following:

1. Initialize the hard- and firmware of the safety part and set all outputs to the safe state
2. perform startup test.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

11.1.2 Parameter State

SRIO-1419 - Parameter State

There shall be a state which shall do the parameter handling of the safety part.

This includes also parameter validation.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

11.1.3 Operating State

SRIO-1420 - Safety Operating State

There shall be a operating state in which the safe outputs and inputs shall be operational, including diagnosis.

Safe fieldbus communication shall be running.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-2089 - Error Handling Safe IO

Error handling in case of a safety relevant failure at digital IO shall be implemented.

A Power-On-Reset and following internal diagnosis shall release the IO in case the failure is not detected again.

Safety relevant errors are e.g. For detected shorts at the IOs, Over /-Under-Load/Voltage & Over-Temperature.

In case a failure is detected at individual IO ports, the error state shall be acknowledged by the customer to set the IO to operate.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

11.1.4 FATAL_ERROR State

SRIO-1421 - ERROR State

There shall be an error state.

Note:

In the state ERROR all outputs shall remain switched off. There shall be no safe fieldbus communication. There shall be no communication between the safety part and the communication part. Whenever the safety part detects a fatal error (e.g. RAM error) it shall switch to ERROR and stop IPC communication. The only way out of this state shall be a reset of the safety controllers.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

11.1.5 FW_UPDATE State

SRIO-2022 - FW_Update State

There shall be a FW-Update state which shall handle the firmware update process.

❗ Must Have, ✅ Approved, ➡ SRIO platform, ⚠ AL400S, ⚠ AL401S, 💰 SIL 3

12 Timing Constraints

moved to system architecture chapter 6 ☐ SRIO-10789

13 Safe Parameterization

13.1 Introduction

The safety relevant parameters are classified into the following types:

1. safety application parameters: (user visible/changeable) parameters that determine the behaviour of the safety application, e.g. filter size of a safe digital input.
2. safe fieldbus parameters: (user visible/changeable) parameters of the safe fieldbus communication, e.g. safe fieldbus address or safe fieldbus watch dog time.

The product supports the following parameterization ways for the safety application parameters:

1. via fieldbus/safe fieldbus: the safety application parameters are sent from the fieldbus master (typically the PLC) to the product, e.g. during fieldbus/safe fieldbus connection establishment.
The safety application uses the safety application parameters that it has received from the fieldbus/safe fieldbus master.
Typically the user does the parameterization with the fieldbus/safe fieldbus engineering tool, e.g. a PLC engineering tool.
2. external Tool: Skipped for actual Projekt: 153829 Remote_IO_Safe_DIDO_PROFIsafe

For using the correct parameters (i.e. corresponding to the correct parameterization way) the parameterization way is selectable via the configuration of the fieldbus/safe fieldbus.

13.2 Safety Application Parameters

Note: In PROFIsafe the safety application parameters are referred to as "iPars".

13.2.1 Safe Inputs

The parameters of the safe inputs are specified in  [SRIO-1157 - Parameters Safe Input](#).

13.2.2 Safe Outputs

The parameters of the safe outputs are specified in  [SRIO-1252 - Parameters Safe Output](#).

13.2.3 Parameter: "Safe Data Mapping"

The term "safe data mapping" refers to the mapping of the safe input and the safe output signals to the safe process data that is transmitted via the safe fieldbus.

13.2.4 Parameter: "Parameterization Way"

The parameter "Parameterization Way" is of relevance only for:

1. the safety application parameters
2. the selection which safety application parameters shall be used: the ones that were written via an external tool or the ones that are sent via the fieldbus/safe fieldbus.

SRIO-1983 - Parameterization Way Via Fieldbus/Safe Fieldbus

The product shall use the safety application parameters that it receives from the fieldbus/safe-fieldbus.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

13.3 Safe Fieldbus Parameters

The term "safe fieldbus parameters" refers to parameters of the safe fieldbus communication, e.g. safe slave's safe fieldbus address or safe fieldbus watch dog time (in PROFIsafe the safe fieldbus parameters are referred to as F-Parameters). The safe fieldbus parameters are described in  [SRIO-688 - 130_Communication](#).

13.4 Own Safe Address

This chapter explains the term "own safe address". The details about setting and using the own safe address are described in more detail in  [130_Communication](#).

In a safe fieldbus communication the communication partners identify each other via their safe addresses. When the safe fieldbus master generates a safe fieldbus telegram the safe fieldbus master codes the safe address of the intended recipient into the safe telegram. When the safe fieldbus slave receives a safe telegram the safe fieldbus slave checks whether the telegram is really meant for him. For this check the safe fieldbus slave needs the following information:

1. who is the intended recipient of the telegram? This information is contained in the received telegram.
2. what is my own safe address? Typical solutions for providing this information to the safe fieldbus slave are:
 - own safe address is stored in non-volatile memory in the safe fieldbus slave
 - own safe address is coded via rotary switches or a DIL switch of the safe fieldbus slave

Please note the difference between the safe fieldbus parameter "slave's safe fieldbus address" (chapter  [SRIO-1952 - Safe Fieldbus Parameters](#)) and "own safe address".

14 Safety related requirements

Some requirements are especially related to the safe functionality.
This include hardware, firmware and regulatory requirements.

SRIO-1777 moved to chapter 1.6 Norms and Standards

SRIO-1778 moved to chapter 1.6 Norms and Standards

SRIO-706 - Mission Time

The mission time must be 20 years.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-1812 - Reboot interval

The needed reboot interval for the device has to be max. 1 Year.

Note: The reboot enables the device to check safety related functions, which cannot be tested while running (e.g. watchdog triggering).

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

SRIO-1961 - Target PFHd

The targeted percentage of PFHd shall be smaller than 1,3 E-8 [1/h].

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

14.1 Intended Use

The device is intended for connecting 1 or 2-channel safe sensors and for connecting suitable 1 or 2-channel safe actuators.

SRIO-1964 - Functionality Safe Input

The device must read signals from connected sensors and transmit the state via the bus system.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-1966 - Functionality Safe Output

The device must receive commanded output states and switch off the corresponding output.

Regardless of the output type selected, this function must comply with SIL3 / Ple cat. 4.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-704 - Reactiontime

The device reaction time should be <100ms.

Remark: The reaction time is the time from:

- input change detected at a F-DI until the input information is transmitted over fieldbus
- or
- the change in a control bit received via fieldbus until the output state is applied.

 Should Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-6566 - Non Persistent Safe State

In case of a Power-Cycle, the device shall release a previously existing safe state.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-6567 - Port Error

In case of an error, the device shall be able to set the related port(s) in safe state.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

15 Environmental conditions

Based on the certification requirements, the environmental conditions to be fulfilled are defined (see also [SRIO-1277 - 180_Housing](#)). The required Norms and Standard are listed here directly or under [1.6 - Norms and standards](#).

15.1 Climatic conditions

SRIO-1800 - Altitude

The device shall be usable at an altitude of 4000m [MSL]

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-1799 - Ambient and storage temperatures

The device shall be usable at Ambient temperature: -40 - 60°C @ 4000 m

Storage temperature: -40 - 80°C

Remark: This must be confirmed according to the environmental test in IEC 61131-2:2017 (IEC 60068-2-2 & IEC 60068-2-14).

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-1256 - Temperature Range

The temperature range of the components on the PCB shall be at least -40°C ... +105°C.

Remark1: Must be confirmed by a data sheet research.

Remark2: If a component has only a range till 85°C, the selection shall be reasoned.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-11808 - Technical specifications for power dependence on ambient temperature

The Hardware shall provide derating specifications for power usage on Inputs or Outputs during high environmental temperatures.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-11930 - Dry Heat

The device shall be subjected to dry heat as defined by IEC 61131-2:2017 [Chapter 5.2].

Remark: Reference check IEC 60068-2-2, Bb

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-11929 - Cold

The device shall be subjected to cold as defined by IEC 61131-2:2017 [Chapter 5.2].

Remark: Reference check IEC 60068-2-1, Test Ab

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-11928 - Temperature cycle resistance test

The device shall be subjected to temperature cycle (resistance) as defined by IEC 61131-2:2017 [Chapter 5.2].

Remark: Reference check IEC 60068-2-14, Test Na

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-11927 - Temperature cycle Insensitivity test

The device shall be subjected to temperature cycle (insensitivity) as defined by IEC 61131-2:2017 [Chapter 5.2].

Remark: Reference check IEC 60068-2-14, Test Nb

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-11926 - Cycles with moist heat

The device shall be subjected to cycles with moist heat as defined by IEC 61131-2:2017 [Chapter 5.2].

Remark: Reference check IEC 60068-2-30, Test Db

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-21369 - EMC Zone

SRIO shall be considered to be used in emc zone B as defined in section 7.1 / table 35 in EN 61131-2:2017.

 Must Have,  Approved,  AL400S,  AL401S,  QM

15.2 Mechanical environmental

SRIO-1781 - The safe Remote IO shall comply with the EN 61010-1 in respect to the following points

The safe Remote IO shall comply with the EN 61010-1:2010 + A1:2019 + A1:2019/AC:2019

- clearance & creepage distance
- fire spread in case of a short circuit within the Device

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-7304 - Resistance of the identification plate

The safe Remote IO shall comply with the EN 61010-1 regarding resistance of the identification plate.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-1780 - The safe Remote IO-hardware shall comply with 61131-2

The safe Remote IO shall comply with the IEC 61131-2:2017 regarding the mechanical resistance (Chapter 5.3).

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-7305 - The safe Remote IO-hardware shall comply with EN 61131-6 regarding mechanical test

The safe Remote IO shall comply with the EN 61131-6:2012 [chapter 12.4] regarding the mechanical requirements.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  QM

SRIO-11813 - Mechanical Shock

The device shall be subjected to mechanical shocks, as defined by IEC 61131-2:2017 [Chapter 5.3].

Remark: Reference check IEC 60068-2-27, Test Ea

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-2037 - DIN EN 60068-2-6/64

The enclosure must provide mechanical protection of the electronic inside.

Enclosure must withstand vibration test according to DIN EN 60068-2-6 and DIN EN 60068-2-64 with typical severity level (transport stress) of 10..500 Hz, 1,87 gRMS for 90 minutes each axis (broadband random vibration) and of 10..150 Hz, 2g for 20 cycles each axis (swept sine) as well as.

The DUT shall remain fully functional during and after the tests and shall issue no error messages.

The tests should be done with complete electronics. During the test, the electronics must be in use and their state of preservation must be monitored.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-9403 - Vibration and temperature profiles

The information on the vibration and temperature profiles shall be given in the datasheet.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

SRIO-11810 - Assembly

The device shall be mounted on a flat surface.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-11809 - Torsional forces or mechanical loads

Torsional forces or mechanical loads during or after installation by the user are excluded and need not be assumed.

Remark: Add note to manual.

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-11910 - Freefall Resistance Test

While packaged, the device shall resist damage after a free fall as defined by IEC 61131-2:2017 [Chapter 5.4.4].

 Must Have,  Approved,  AL400S,  AL401S,  QM

SRIO-3784 - Drop test

The device falls down on the floor during storage, transport, mounting or general handling. The unit must withstand this without damage or damage must be visible.

- The maximum permissible height of fall is to be assumed as 1.00 m.
- withstand drop test according to IEC 60068-2-31

 Must Have,  Approved,  AL400S,  AL401S,  QM

15.2.1 IP Protection

see chapter 3  SRIO-1287 - IP Protection Following in  180_Housing

16 EMC Requirements

The required Norms and Standart are listed here directly or under [1.6 - Norms and standards](#).

SRIO-1779 - The safe Remote IO shall comply with the EN 61326-3-1

The safe Remote IO shall comply with the EN 61326-3-1:2017 regarding the EMC (safety related levels)

🔴 Must Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 🟡 SIL 3

SRIO-7306 - The safe Remote IO-hardware shall comply with EN 61131-2 regarding the EMC resistance

The safe Remote IO shall comply with the EN 61131-2:2007 regarding the EMC resistance.

🔴 Must Have, ✅ Approved, ➡ SRIO platform, 📌 AL400S, 📌 AL401S, 🟡 QM

SRIO-20174 - Connection Cabel: Restriction Shielding

Following restriction, regarding shielding of the cable for each interface, shall apply:

Interface	Categorization	Shielding [according to 61131-2]
XF1, XF2	Data communication	yes (grounded at both ends)
XD1, XD2	Power supply	no
F-DI Port	Functional safety Digital Inputs	no
F-DO Port	Functional safety Digital Outputs	no

🔴 Must Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 🟡 QM

SRIO-20173 - Functional earth connection: Restriction Cable Length

The length of the FE connection to the device shall be < 3 meter.

🔴 Must Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 🟡 QM

SRIO-9841 - Restriction Inhouse

The usage of the device and all belonging cable installation is restricted to indoor installation only.

🔴 Must Have, ✅ Approved, ➡ SRIO platform, 📌 AL400S, 📌 AL401S, 🟡 QM

SRIO-20619 - Immunity Industrial Environments

SRIO shall fulfill the related requirements according to radiation immunity as specified in IEC 61000-6-2:2019.

🔴 Must Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 🟡 QM

SRIO-9043 - Radiation Emission

SRIO shall fulfill the related requirements according to radiation emission as specified in IEC 61000-6-4:2019.

🔴 Must Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 🟡 QM

16.1 Assesment Criterion

16.1.1 Criterion A

SRIO-11573 - Criterion A

The safe remote IO works as intended if the device works during and after the tests uninterrupted.

Interface:	Supervision:	Definition:
PROFINET Communication	Test with a connected safety PLC (S7). This plc shall supervise the connection and log the potential connection interrupts.	During and after the tests: No interrupts to connection.
PROFIsafe Communication	Test with a connected safety PLC (S7). Connection and I/Os shall be supervised. An external stimulation of the safety inputs shall be implemented. The outputs of the system shall be connected to plc inputs to verify a statechange.	During and after the tests: No interrupts to connection. No uncontrolled change of the I/O signals.
IoT Core Communication	Not supervised	After the tests: <i>IoT Core access possible.</i>
FE- Anschluss	Not supervised	

Safe Inputs	External stimulation of the safety inputs, verified by connected safety plc (S7). Stimulation in the way to have a mix of static high, static low and toggling for input signals.	During and after the tests: No uncontrolled change of the I/O signals.
Safe Outputs	The outputs of the system shall be connected ohmic loads. The output signal shall be connected to plc inputs to verify a state change. Stimulation in the way to have a mix of static high, static low and toggling for output signals.	During and after the tests: No uncontrolled change of the I/O signals.
LED	Supervision of P- and FS-LED regarding color and constancy. The other LEDs are not supervised.	During and after the tests: P-LED shall be static green FS-LED shall be off [no supervision needed, impression of tester is sufficient]

Remark: The appliance (equipment, device) must continue to work as intended during and after the test.

🚫 Must Have, ✅ Approved, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3

16.1.2 Criterion B

SRIO-11572 - Criterion B

The test is passed if the device works after the tests uninterrupted.

In difference to Criterion A disturbances during the test is allowed as far as there is no change in the operationstate.

Interface:	Supervision:	Definition:
PROFINET Communication	Test with a connected safety PLC (S7). This plc shall supervise the connection and log the potential connection interrupts.	During tests: Disturbances e.g. frame drop tolerated. After the tests: Automatically reconnects in between tests or at least at the end of the test to the plc.
PROFIsafe Communication	Test with a connected safety PLC (S7). Connection and I/Os shall be supervised. An external stimulation of the safety inputs shall be implemented. The outputs of the system shall be connected to plc inputs to verify a statechange.	During tests: Interruption of communication is tolerated. After the tests: Restart of communication by acknowledgment.
IoT Core Communication	Not supervised	After the tests: <i>IoT Core access possible.</i>
FE- Anschluss	Not supervised	
Safe Inputs	External stimulation of the safety inputs, verified by connected safety plc (S7). Stimulation in the way to have a mix of static high, static low and toggling for input signals.	During tests: LEDs might flicker [no supervision needed, impression of tester is sufficient] Process data with Qualifier exchanged or After the tests: with acknowledgement correct process data with Qualifier = good resumed.
Safe Outputs	The outputs of the system shall be connected ohmic loads. The output signal shall be connected to plc inputs to verify a state change. Stimulation in the way to have a mix of static high, static low and toggling for output signals.	During tests: LEDs might flicker [no supervision needed, impression of tester is sufficient] Process data applied or After the tests: with acknowledgment output of process data resumed
LED	Supervision of P- and FS-LED regarding color and constancy. The other LEDs are not supervised.	During tests: LEDs might flicker [no supervision needed, impression of tester is sufficient] After the tests: P-LED shall be static green FS-LED shall be off

Remark: The appliance (equipment, device) must continue to work as intended after the test.

🚫 Must Have, ✅ Approved, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3

16.1.3 Criterion DS

SRIO-11574 - Criterion DS

During the test: Functions of the EUT that are intended for safety applications:

- 1) As assessment criterion A, B or
- 2) Can be permanently or temporarily disrupted if the EUT reacts to a fault in such a way that it can be detected and the EUT reaches a defined state and remains there.

Functions that are not intended for safety applications can be temporarily or permanently disrupted.

After the test: Defined state may be solved with a power cycle. Destruction of safety-related components is permitted if the defined state of the EUT can be maintained. Destruction of non-safety-related components is permitted.

 Must Have,  Approved,  AL400S,  AL401S,  SIL 3

16.2 EMC Immunity during different systemstates

SRIO has different safety substate the criterion defined in [16.1 - Assessment Criterion](#) define the behavior during [11.1.3 - Operating State](#). In the other substates, there is no active configuration and therefore also no active safety communication / cyclic data exchange. In those substates the device has either I/Os deactivated (e.g. [11.1.2 - Parameter State](#) or [11.1.5 - FW_UPDATE State](#)) or is in a safe state e.g. [11.1.4 - FATAL_ERROR State](#).

SRIO-11987 - EMC Immunity during substates excluding operate

During all states excluding OPERATING, I/Os shall remain deactivated (safe state).

Remark: During emc disturbance the device is allowed to switch to safe state during emc influence depending on criterium B.

 Must Have,  Approved,  SRIO platform,  AL400S,  AL401S,  SIL 3

16.3 FCC and ICES

SRIO-11937 - FCC and ICES EMV Requirements

The device shall fulfill the emc requirements regarding Class A defined by FCC [USA] and ICES [CAN].

 Must Have,  Approved,  AL400S,  AL401S,  QM

17 Test Related Requirements

To fulfill the required production test, appropriate features (e.g. test-points or SW-interfaces) must be included in the design. This chapter lists the test requirements.

17.1 Functional Test

SRIO-1936 - HW/SW integration tests

Requirements to perform HW/SW integration tests must be considered, therefore a appropriate document must be created.

❗ Must Have, ✅ Approved, ↔ SRIO platform, 📌 AL400S, 📌 AL401S, 💰 SIL 3

SRIO-6580 - Firmware Tests

The tests shall test the functionality of the components by using firmware functions.

❗ Must Have, ✅ Approved, ↔ SRIO platform, 📌 AL400S, 📌 AL401S, 💰 SIL 3

SRIO-6582 - Test expectation

The produced device shall be tested to confirm all relevant components/functions are working.

❗ Must Have, ✅ Approved, ↔ SRIO platform, 📌 AL400S, 📌 AL401S, 💰 SIL 3

SRIO-2096 - The safe Remote-IO-Profisafe Device shall be supported by a F-CPU

The following PLC-Types should be supported:

- Siemens
 - S7-12nn (with F-CPU)
 - S7-15nn (with F-CPU) (1511F-1PN will be focus for testing)
 - S7-300-F
- Phoenix
 - RFC407S PLCWorx (nice to have)
 - RFC4072S PLCnext (mandatory)

Remark:

During the TestPlan-creation we decided to use the additional PLCs / FCPU's as well.

For several big customers it is mandatory to use the P6x PLCs

Siemens PLCs shall be used as the standard PLCs, Phoenix PLCs, shall be used as interop-clarification... .

❗ Should Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 📌 QM

SRIO-6599 - Support Siemens PLC

The device shall support the communication to 1511F-1PN.

❗ Must Have, ✅ Approved, ↔ SRIO platform, 📌 AL400S, 📌 AL401S, 📌 QM

SRIO-6600 - Interoperability Siemens PLC

The device shall be interoperabel to S7-300-F.

❗ Must Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 📌 QM

SRIO-6598 - Interoperability Phoenix PLC

The device shall be interoperabel to RFC4072S with PLCnext.

❗ Must Have, ✅ Approved, 📌 AL400S, 📌 AL401S, 📌 QM

17.2 Production related tests

Requirements from manufacturing to perform production testing must be considered.

This requirement is a placeholder for requirements coming from IFM production and is working in progress.

The Manufacturing - Test is divided into several parts:

- 1.) Testing on a LPB / LPK (PCB after SMD-process) --> ICT, visual inspection & intermediate test (during the intermediate test, the functionality of the later product will be tested in respect to the EA-System, min / max. tolerances of the circuit etc. --> definition what had to be tested shall be provided by the TestMgm
- 2.) The produced and "pre-tested" LPB will be send from ifm electronic (placed in TT--> division ZVL) to ifm ecolink
- 3.) ifm ecolink shall perform a additional test, to secure that no failures appeared during transport of the LPBs
- 4.) Test after firmware-upload
- 5.) Test after/during the injection process
- 6.) End-Test

17.2.1 Tett nang

SRIO-6579 - Testcoverage ZVL

As defined by the test specification, the required test shall be performed.

🚫 Must Have, ✅ Approved, ➡️ SRIO platform, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3

SRIO-6578 - PCB Tests

The tests shall verify that all components are correct, available, orientation and polarity is ok and properly sautered.

🚫 Must Have, ✅ Approved, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3

SRIO-6584 - DMC

Each printed circuit board shall have a data matrix code, identifying the pcb, lasered on the LED side.

🚫 Must Have, ✅ Approved, ➡️ SRIO platform, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3

SRIO-1795 - An intermediate - test has to be performed at ZVL-TT

An intermediate - test has to be performed at ZVL-TT* (for outgoing goods)

The aim of this test is a functional test of the LPBs (this might be implemented for not separated PCB-Panels)

Remark: has to be a part of the test-spec

(* ZVL-TT stands for Zentrale Vorfertigung Leiterplatten in Tett nang)

🚫 Must Have, ✅ Approved, ➡️ SRIO platform, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3

17.2.2 Ecolink

SRIO-6581 - Testcoverage ecolink

As defined by the test specification, the required test shall be performed.

🚫 Must Have, ✅ Approved, ➡️ SRIO platform, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3

SRIO-1796 - An intermediate - test has to be performed at ecolink

An intermediate - test has to be performed at ecolink (for incoming goods)

The standard approach might be reused therefore

Reason for the intermediate test:

- for a 100% traceability, we need to store the production information at the device (LPB, order Nr. / Article-Production-Code etc.) this shall be done at the different production lvl.
- The LPB shall be tested with a high degree of test-coverage to ensure, that in case a failure appears we are able to repair the LPBs. (after injection, that is no longer possible)

🚫 Must Have, ✅ Approved, ➡️ SRIO platform, ⚠️ AL400S, ⚠️ AL401S, 📄 QM

SRIO-1797 - The finished Products have to be tested (100%-pcs. Test) at a defined Endtest

The finished Products have to be tested (100%-pcs. Test, every unit has to be tested) at a defined Endtest.

🚫 Must Have, ✅ Approved, ➡️ SRIO platform, ⚠️ AL400S, ⚠️ AL401S, 💰 SIL 3